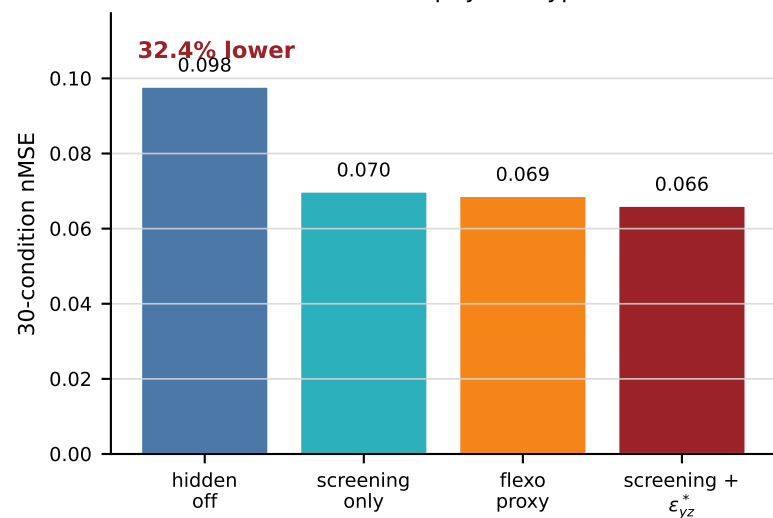


Residual-guided hidden-physics discovery: screening plus localized anisotropic eigenstrain

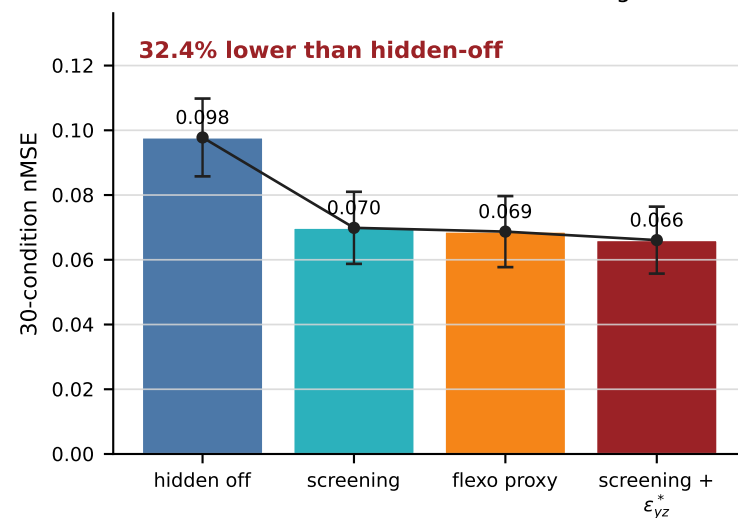
a

candidate hidden-physics hypotheses



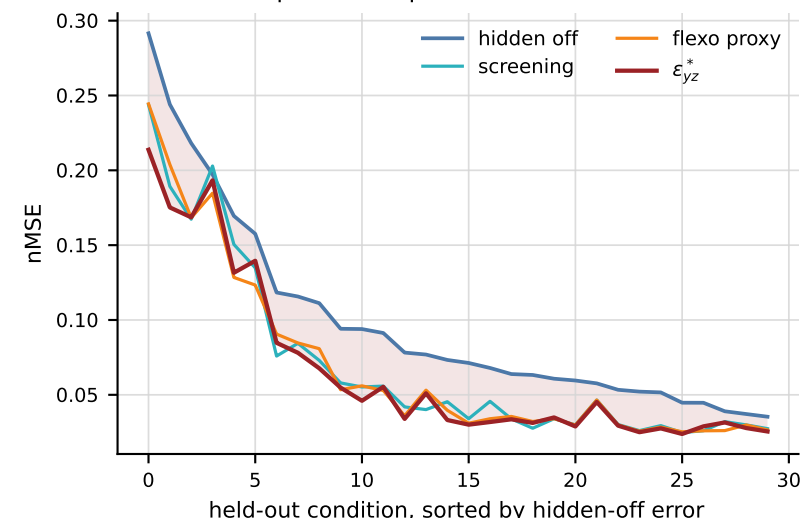
b

full holdout confirms the ordering



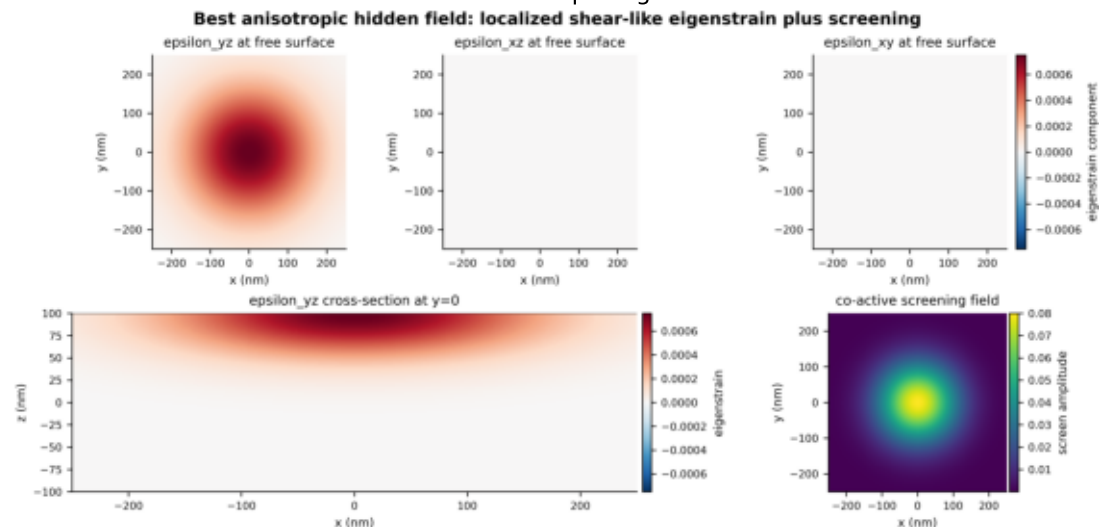
c

anisotropic field improves all 30 holdout cases



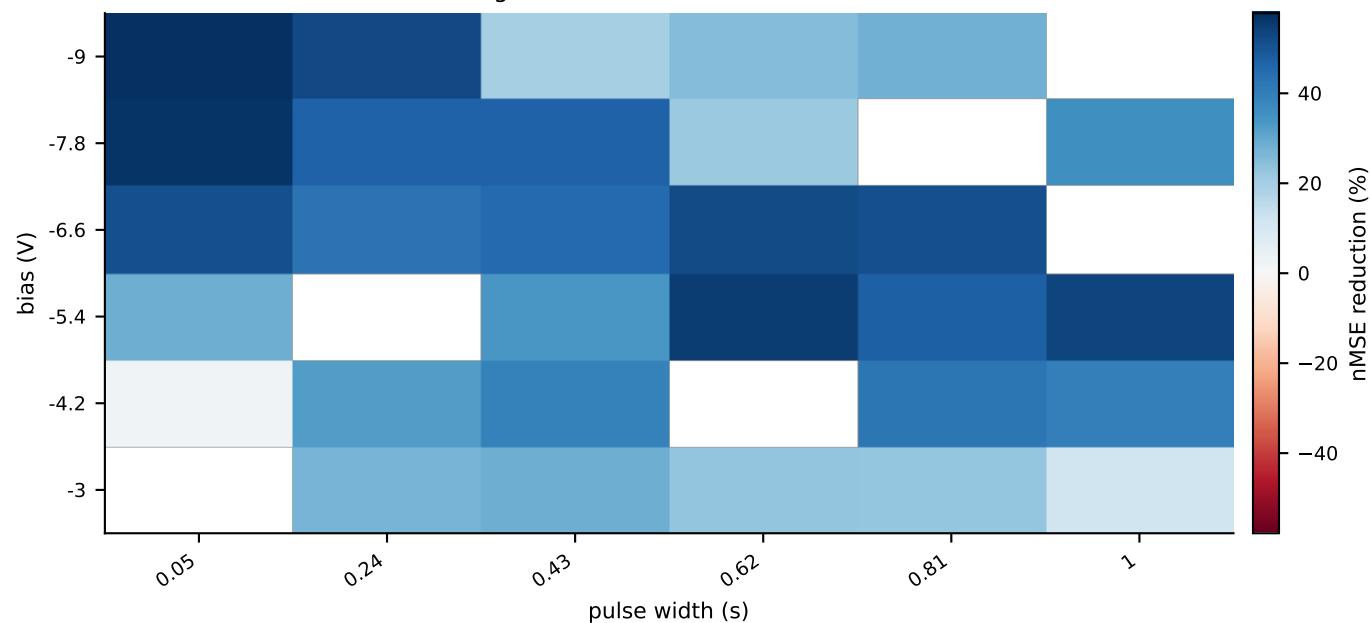
d

best low-rank anisotropic eigenstrain field



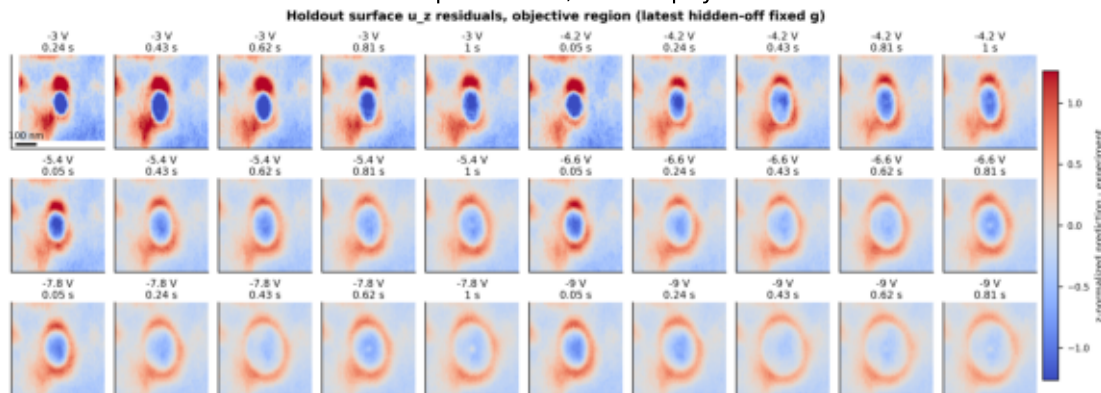
e

gain over hidden-off baseline



f

center-crop residuals, hidden physics off



g

center-crop residuals, screening + ϵ_{yz}^*

